MTH209: Endsem Exam - Question 1

1. Follow the instructions **exactly**.
2. If your code is not properly commented or horizontal/vertical spacing is missing, points will be deducted.
3. In this question, you are NOT allowed to use any external libraries.

Question

The goal of this question is to compare the mean squared error of OLS, Ridge, and Lasso estimators. Consider the following linear regression model:

$$y \sim N\left(X\beta^*, \sigma^2 \mathbb{I}_n\right) ,$$

where $y \in \mathbb{R}^n$, $X \in \mathbb{R}^{n \times p}$, $\beta^* \in \mathbb{R}^p$ and $\sigma^2 > 0$. You may use the following simulation settings to test your code.

```
set.seed(209)
n <- 50
p <- 2
beta.star <- c(-5, .1)
sigma2 <- 1

# Making design matrix
X <- matrix(rnorm(n*p), nrow = n, ncol = p)
```

Functions to submit

1. Write a function `ridge` that takes input `y`, `X`, and `lambda` and returns the ridge regression estimator:

```
# X: n x p matrix of covariates
# y: a vector of length n
# lambda: a numeric value
ridge <- function(y, X, lambda)
{
  ...
  est <- ...
  return(est)
}
```

2. Write a function `lasso` that takes input `y`, `X`, and `lambda` and returns the lasso regression estimator:

```
# X: n x p matrix of covariates
# y: a vector of length n
# lambda: a numeric value
lasso <- function(y, X, lambda)
```

```
{
  ...
  est <- ...
  return(est)
}
```

3. Write a function `mse_regression` that returns the Mean Squared Error of the OLS, Ridge and Lasso regression estimators.

```
# X: n x p matrix of covariates
# beta.star: a numeric vector of length p of true coefficents
# sigma2: sigma^2 variance of the responses
# reps: number of replications
# lambda: a numeric value
mse_regression <- function(X, beta.star, sigma2, reps, lambda)
{


  ...
  for(r in 1:reps)
  {
  ...
  ...
  }

  # resuts MUST be in this order
  names(mse_beta) <- c("OLS", "Ridge", "Lasso")

  # final MSE must be stored in a vector of length 3 called mse_beta
  mse_beta <- ...
  return(mse_beta)
}
```

Submission Instructions

INSTRUCTIONS: copy and paste ALL THREE functions into a file called `question1.R` and upload it to mooKIT. Upload **ONLY the function** and nothing else. This question has 35 marks.